Chapter 2 — Variables, Types and Operators

Programs store values and compute new ones. This chapter covers **variables** (names for values), **types** (what kind of value something is) and **operators** (`+`, `===`, `&&`, `??` …), plus **type coercion** — the automatic conversion of one type into another — so that `"3" + 4`, `0 == ""` and `0.1 + 0.2` stop being mysteries.

What you'll learn: `let` / `const` / `var`; hoisting and the TDZ; the seven primitives, objects and `typeof`; value vs reference; arithmetic, comparison, logical, `??`, `?.`, bitwise and assignment operators and precedence; `==` vs `===`; numbers in depth (precision, `NaN`, `Infinity`, safe integers, `Number` API, `BigInt`); truthy/falsy and explicit conversions.

Code: `code/` — every "Output:" block is real Node 24 output. Files are plain `.js` (sloppy-mode scripts) unless noted (Chapter 1 §1.8).

2.1 Variables: `let`, `const` and `var`

What it is

A **variable** is a named storage slot for a value. **Declare** creates the name (`let score;`), **assign** stores a value (`score = 10;`), **initialise** does both (`let score = 10;`). A declared-but-unassigned variable holds `undefined`. A `const` is a label glued to one value: you can't swap it, but if the value is an object you can still change what's inside.

How it works — the three keywords

```
// code/01_declarations.js
let score;                    // declared, no value yet
console.log("score before assignment:", score);
score = 10;                   // assign later
score = score + 5;            // reassign (let allows it)
console.log("score:", score);

const PI_APPROX = 3.14;       // const must be initialised and can't be reassigned
try { eval("PI_APPROX = 3"); } catch (e) { console.log("reassign const ->", e.name + ":", e.message); }

// const protects the BINDING, not the object: the contents can still change.
const cart = { items: 1 };
cart.items = 2;               // allowed: we mutate the object, not the variable
console.log("const object mutated:", cart);

// var: the old way. Function-scoped, can be redeclared, hoisted with value undefined.
var legacy = "first";
var legacy = "second";        // redeclaring with var is silently allowed
console.log("var redeclared:", legacy);
try { eval("let dup = 1; let dup = 2;"); } catch (e) { console.log("let redeclared ->", e.name); }

let a = 1, b = 2, c = a + b;  // several declarations in one statement
console.log("a, b, c =", a, b, c);

const greeting = "Hello";
const who = "World";
console.log(greeting + " " + who);
```

Output:

```
score before assignment: undefined
score: 15
reassign const -> TypeError: Assignment to constant variable.
const object mutated: { items: 2 }
var redeclared: second
let redeclared -> SyntaxError
a, b, c = 1 2 3
Hello World
```

(`eval` runs a string as code — used here only to catch a deliberate error. Don't use it in real code.)

Comparison table

	`let`	`const`	`var`
Introduced	ES2015	ES2015	original (1995)
Scope (where the name is visible)	**`block { }`**	**`block { }`**	**`function`** (or global)
Reassign	yes	**no** (`TypeError`)	yes
Redeclare in same scope	no (`SyntaxError`)	no	**yes, silently**
Must initialise	no	**yes**	no
Use before the declaration line	`ReferenceError` (TDZ, §2.2)	`ReferenceError` (TDZ)	gives `undefined`
At top level of a classic script, becomes a property of `globalThis`	no	no	**yes**

Scope is covered in Chapter 4 §4.5; in short, `let`/`const` inside `{ }` exist only inside those braces, `var` leaks to the whole function.

Which one should I use?

1. **`const` by default** — tells the reader the name never re-points.
2. **`let` when the value must change** — counters, accumulators, branch assignments.
3. **`var` never in new code** — recognise it in old code; its function scope, silent redeclaration and hoisting cause the bugs `let`/`const` prevent.

Naming rules and conventions

- Letters, digits, `_`, `$`; not starting with a digit. Unicode letters work (`café`) but stick to ASCII in shared code. Case-sensitive. Reserved words (`let`, `class`, `return`, `typeof` …) are not allowed.
- **`camelCase`** for variables/functions (`totalPrice`), **`PascalCase`** for classes (`ShoppingCart`, Chapter 9), **`UPPER_SNAKE_CASE`** for fixed config constants (`MAX_RETRIES`). Booleans as questions: `isOpen`, `hasAccess`.

Common mistakes

- **`const` ≠ read-only object** — it only blocks reassignment; use `Object.freeze` (Chapter 5).
- **Forgetting the keyword** — `total = 5` creates an accidental global in sloppy mode, throws in strict mode.
- **Using a block's `let` outside the block** → `ReferenceError`.

TL;DR

`const` by default, `let` to reassign, `var` only when reading old code. `const` freezes the **binding** (the name→value link), not the value.

2.2 Hoisting and the Temporal Dead Zone

What it is

Before running a scope, the engine creates all its declared names — **hoisting**. What differs is the value before the declaration line runs:

Declaration	Before its line runs
`var x`	exists, value `undefined`
`let x`, `const x`, `class X`	exists but **uninitialised** — touching it throws `ReferenceError`
`function f() {}`	exists and is **fully usable** (covered in Chapter 4)

The window between entering the scope and a `let`/`const` declaration line is the **Temporal Dead Zone (TDZ)**.

Example

```
// code/02_tdz_and_hoisting.js
console.log("var before its line:", typeof hoistedVar, hoistedVar); // declared + set to undefined at start
var hoistedVar = "ready";

try {
  console.log(notYet);        // let exists but is uninitialised until its line runs
} catch (e) {
  console.log("let before its line ->", e.name + ":", e.message);
}
let notYet = "ready";

try { typeof inZone; let inZone = 1; } catch (e) { console.log("typeof in TDZ ->", e.name); }
{
  // The TDZ is about TIME, not position: a function may mention a later let,
  // as long as it is called after the let line has run.
  const show = () => later;
  let later = "defined by the time show() is called";
  console.log(show());
}
```

Output:

```
var before its line: undefined undefined
let before its line -> ReferenceError: Cannot access 'notYet' before initialization
typeof in TDZ -> ReferenceError
defined by the time show() is called
```

Why it matters

With `var`, early reads silently give `undefined` and the bug surfaces elsewhere; `let`/`const` fail immediately with a named error. Even `typeof` — normally safe on unknown names — throws in the TDZ.

In the interview

"What is the Temporal Dead Zone?" `let`, `const` and `class` are hoisted — the scope knows them from the start — but not initialised until their declaration is evaluated; access in that window throws `ReferenceError`. `var` is hoisted *and* initialised to `undefined`.

- *Are `let`/`const` hoisted?* Yes — that's why an inner `let x` **shadows** (hides) an outer `x` even on lines above it: you get a TDZ error, not the outer value.
- *Lines or time?* Time: a closure (a function remembering surrounding variables — Chapter 4) referring to a later `let` is fine if it runs after the declaration.

TL;DR

All declarations are hoisted; `var` starts as `undefined`, `let`/`const` start in the TDZ and throw if touched early.

2.3 Data types and `typeof`

What it is

JavaScript has **eight** types: seven **primitives** and **object**. It is **dynamically typed**: values have types, variables don't.

Type	Example values	Notes
`number`	`42`, `3.14`, `-0`, `Infinity`, `NaN`	One type for integers and decimals: 64-bit floating point (§2.10)
`bigint`	`10n`, `2n ** 64n`	Whole numbers of any size (ES2020)
`string`	`"hi"`, `'hi'`, `` `hi ${x}` ``	Text, immutable (Chapter 6)
`boolean`	`true`, `false`	§2.11
`undefined`	`undefined`	"No value assigned yet" — what JS itself gives you
`null`	`null`	"Intentionally empty" — what *you* assign to mean "nothing"
`symbol`	`Symbol("id")`	Unique identifiers, mostly for object keys (Chapter 8)
`object`	`{ a: 1 }`, `[1, 2]`, functions, dates, maps …	Everything that isn't a primitive

`typeof` — asking a value its type

```
// code/03_types_typeof.js (abridged)
for (const [label, value] of samples) {   // samples: [["number", 42], ["null", null], ["array", [1, 2]], ...]
  console.log(label.padEnd(15), "typeof ->", typeof value);
}
```

Output:

```
number          typeof -> number
number (float)  typeof -> number
NaN             typeof -> number
bigint          typeof -> bigint
string          typeof -> string
boolean         typeof -> boolean
undefined       typeof -> undefined
null            typeof -> object
symbol          typeof -> symbol
object          typeof -> object
array           typeof -> object
function        typeof -> function
date            typeof -> object
Array.isArray([1,2]) -> true
value === null      -> true
box: number
box: string
```

Quirks to memorise:

1. **`typeof null === "object"`** — a first-implementation bug that can never be fixed. Test with `value === null`.
2. **`typeof NaN === "number"`** — NaN is a special number value.

3. **Functions give `"function"`** (though they're objects); arrays give `"object"` — use `Array.isArray`.

`typeof` always returns a string and is safe on undeclared names (`"undefined"`), except in a TDZ.

`undefined` vs `null`

	`undefined`	`null`
Meaning	value not set	deliberately empty
Who produces it	the language: unassigned variables, missing properties, missing arguments, functions without `return`	your code (and some APIs, e.g. `document.getElementById` when nothing matches)
`typeof`	`"undefined"`	`"object"` (quirk)
`Number(…)`	`NaN`	`0`
`== null`	`true`	`true`

Idiom: `value == null` is true for both `null` and `undefined` and nothing else — the one widely accepted use of `==`.

TL;DR

Seven primitives + objects. `typeof` identifies them, except `null` (→ `"object"`) and arrays (use `Array.isArray`).

2.4 Primitives vs objects

What it is

- A **primitive** is an **immutable** value; the variable holds the value itself; copying copies the value.
- An **object** is a mutable collection of properties; the variable holds a **reference** (an address). Copying copies the reference, so both variables point at the *same* object.

Example

```
// code/04_primitives_vs_objects.js
let x = 10;
let y = x;        // copies the value 10
y = 20;
console.log("x, y:", x, y);

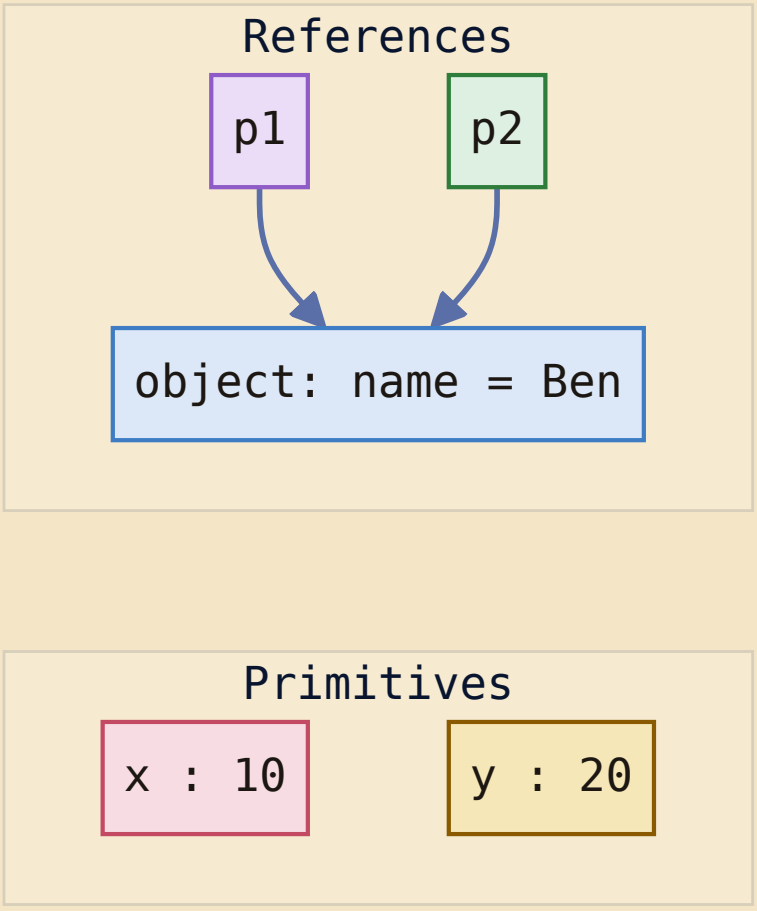
const p1 = { name: "Ana" };
const p2 = p1;    // copies the REFERENCE — both names point at one object
p2.name = "Ben";
console.log("p1.name:", p1.name);

console.log("{} === {}:", {} === {});          // two different objects
console.log("p1 === p2:", p1 === p2);          // same object
console.log("'a' === 'a':", "a" === "a");      // primitives compare by value

const s = "hello";
const upper = s.toUpperCase();                 // string methods return NEW strings
console.log(s, upper);
console.log("length of 'hello':", s.length, "| (255).toString(16):", (255).toString(16));
console.log("typeof new Number(5):", typeof new Number(5), "| typeof Number('5'):", typeof Number("5"));
```

Output:

```
x, y: 10 20
p1.name: Ben
{} === {}: false
p1 === p2: true
'a' === 'a': true
hello HELLO
length of 'hello': 5 | (255).toString(16): ff
typeof new Number(5): object | typeof Number('5'): number
```



Autoboxing and wrapper objects

Primitives have no properties, yet `"hello".length` works: the engine temporarily wraps the primitive in a **wrapper object** (`String`, `Number`, `Boolean`, `BigInt`, `Symbol`), reads the property and discards it — **autoboxing**.

Don't create wrappers with `new Number(5)` / `new String("x")` / `new Boolean(false)`: they're objects (`typeof` → `"object"`, `new Number(5) === 5` is `false`, `new Boolean(false)` is truthy). Call them **without** `new` — `Number("5")`, `String(5)`, `Boolean(0)` — to *convert* to primitives.

Why it matters

- A function receiving an object can change *your* object (Chapter 4).
- `===` on objects means "same object?", not "same contents?".
- React detects changes by checking for a *new* object — hence "don't mutate (change in place) state".

> **TL;DR**
>
> Primitives: immutable, copied/compared by value. Objects: mutable, copied/compared by reference. Convert with `Number()/String()/Boolean()` without `new`.

2.5 Arithmetic operators

What it is

An **operator** takes **operands** and produces a value: unary (`-x`, `!x`), binary (`a + b`), and one ternary (`c ? a : b`, Chapter 3).

Operator	Name	Example	Result
`+`	addition (or string concatenation)	`10 + 5`	`15`
`-`	subtraction	`10 - 5`	`5`
`*`	multiplication	`10 * 5`	`50`
`/`	division (always floating point)	`7 / 2`	`3.5`
`%`	remainder	`10 % 3`	`1`
`**`	exponentiation (ES2016)	`2 ** 10`	`1024`
`++`	increment by 1	`n++` / `++n`	see below
`--`	decrement by 1	`n--` / `--n`	see below
unary `+` / `-`	convert to number / negate	`+"42"`	`42`

Example

```
// code/05_arithmetic.js (abridged)
console.log("7 / 2 =", 7 / 2, "(no integer division)");
console.log("Math.trunc(7 / 2) =", Math.trunc(7 / 2));
console.log("-7 % 3 =", -7 % 3, "(sign follows the left operand)");
console.log("1 / 0 =", 1 / 0, "| -1 / 0 =", -1 / 0, "| 0 / 0 =", 0 / 0);

let n = 5;
console.log("n++ returns", n++, "then n is", n);   // returns old value
console.log("++n returns", ++n, "then n is", n);   // returns new value

console.log('"3" + 4 =', "3" + 4);
console.log('"3" - 4 =', "3" - 4);
console.log('"3" * "4" =', "3" * "4");
console.log('1 + 2 + "3" =', 1 + 2 + "3");
console.log('"1" + 2 + 3 =', "1" + 2 + 3);
console.log('+"42" =', +"42", '| -"5" =', -"5", '| +"" =', +"", '| +"abc" =', +"abc");
```

Output:

```
7 / 2 = 3.5 (no integer division)
Math.trunc(7 / 2) = 3
-7 % 3 = -1 (sign follows the left operand)
1 / 0 = Infinity | -1 / 0 = -Infinity | 0 / 0 = NaN
n++ returns 5 then n is 6
++n returns 7 then n is 7
"3" + 4 = 34
"3" - 4 = -1
"3" * "4" = 12
1 + 2 + "3" = 33
"1" + 2 + 3 = 123
+"42" = 42 | -"5" = -5 | +"" = 0 | +"abc" = NaN
```

How it works — the rules behind the surprises

1. **Division never throws:** `±Infinity` for division by zero; `0 / 0` is `NaN`.
2. **`%` is remainder, not modulo:** sign of the left operand (`-7 % 3` is `-1`). Always-positive modulo: `((a % n) + n) % n`.

3. **Prefix vs postfix:** both change `n`; postfix `n++` *returns* the old value, prefix `++n` the new one. Alone on a line, no difference.
4. **`+` is overloaded:** if either operand (after converting objects to primitives) is a string, it concatenates; otherwise adds. Left to right: `1 + 2 + "3"` → `"33"`, `"1" + 2 + 3` → `"123"`.
5. **Other arithmetic operators always convert to numbers:** `"3" - 4` → `-1`.
6. **Unary `+`** is the shortest number conversion; `""` → `0`, non-numeric text → `NaN`.
7. **`-2 ** 2` is a `SyntaxError`** — write `(-2) ** 2` or `-(2 ** 2)`. `**` is right-associative: `2 ** 3 ** 2` = `2 ** 9` = `512`.

Common mistakes

- Form values are strings: `"2" + "3"` → `"23"`. Convert first: `Number(input.value)`.
- Expecting `7 / 2` to be `3` — use `Math.trunc`/`floor`/`round` (Chapter 5).

2.6 Comparison operators and equality

What it is

Comparisons always produce a **boolean**.

Operator	Meaning	Converts types?
`===`	strictly equal: same type **and** same value	no
`!==`	strictly not equal	no
`==`	loosely equal: equal after converting types	**yes**
`!=`	loosely not equal	**yes**
`>` `<` `>=` `<=`	greater / less (or equal)	yes (to number, unless both are strings)

Example

```js
// code/06_comparison_equality.js (abridged)
const x = 5, y = "5";
console.log("x == y  :", x == y);
console.log("x === y :", x === y);
console.log("x != y  :", x != y);
console.log("x !== y :", x !== y);
console.log("x > 2, x < 10, x >= 5, x <= 5:", x > 2, x < 10, x >= 5, x <= 5);
// ... loops over a table of pairs, printing == and === for each
```

Output:

```
x == y  : true
x === y : false
x != y  : false
x !== y : true
x > 2, x < 10, x >= 5, x <= 5: true true true true
--- loose equality surprises ---
0         == ""        -> true  | === -> false
0         == "0"       -> true  | === -> false
""        == "0"       -> false | === -> false
false     == "0"       -> true  | === -> false
false     == ""        -> true  | === -> false
null      == undefined -> true  | === -> false
null      == 0         -> false | === -> false
undefined == 0         -> false | === -> false
NaN       == NaN       -> false | === -> false
[1]       == 1         -> true  | === -> false
[]        == ""        -> true  | === -> false
[1,2]     == "1,2"     -> true  | === -> false
--- relational comparisons ---
"10" < "9"  : true (both strings: compared character by character)
"10" < 9    : false (one number: string converted to number)
"apple" < "banana": true | "Z" < "a": true
null >= 0   : true | null == 0: false
NaN < 1, NaN >= 1: false false
--- Object.is ---
NaN === NaN: false | Object.is(NaN, NaN): true
0 === -0: true | Object.is(0, -0): false
```

`0 == ""` and `0 == "0"` are true, yet `"" == "0"` is false: loose equality isn't even **transitive** — the main reason to avoid it.

How it works — the `==` algorithm (simplified from the spec)

For `a == b`:

1. **Same type** → compare like `===`.
2. **`null` and `undefined`** equal each other and **nothing else** (`null == 0` is false).
3. **Number vs string** → string to number (`""` → `0`).
4. **Boolean vs anything** → boolean to number first. So `false == "0"` is true, and `if (x == true)` is a trap: `"2" == true` → `2 == 1` → false.
5. **Object vs primitive** → object to primitive (`valueOf`/`toString`; arrays join: `[1,2]` → `"1,2"`, `[]` → `""`), compare again.
6. **BigInt vs number** → compare mathematical values (`1n == 1` is true).
7. **`NaN`** equals nothing, including itself.

`===` skips all that: different types → `false`. Remaining oddities: `NaN === NaN` is `false`, `0 === -0` is `true`; `Object.is(a, b)` is `===` with those two fixed.

Relational comparisons (`<`, `>` …)

1. Objects → primitives.
2. **Both strings** → compared by UTF-16 code unit (JS stores text as 16-bit **code units**, one or two per character) — dictionary order by character code: `"10" < "9"`, and uppercase before lowercase (`"Z" < "a"`). For human sorting use `localeCompare` (Chapter 6).
3. Otherwise both → numbers. `null` → `0`, so `null >= 0` is true though `null == 0` is false. Anything vs `NaN` is false.

Trade-offs — `==` vs `===`

	`===`	`==`
Predictable	yes	needs the rules above
Catches type bugs	yes (`"5" === 5` false)	hides them
Useful idiom	—	`x == null` (null **or** undefined)
Linters (tools that scan code for likely bugs; ESLint rule `eqeqeq`)	recommended	flagged, except `== null` if configured

In the interview

30-second answer — "Difference between `==` and `===`?" `===` compares type and value with no conversion. `==` converts first: null/undefined equal only each other; strings and booleans become numbers; objects become primitives. That makes `==` non-transitive (`0 == ""`, `0 == "0"`, but `"" != "0"`), so I use `===` everywhere except `x == null`.

- *Check for `NaN`?* `Number.isNaN(x)` (or `Object.is(x, NaN)`); `x === NaN` is always false.
- *`[] == ![]`?* `true`: `![]` is `false`, then `[] == false` → `"" == 0` → `0 == 0`.
- *`null >= 0` true but `null == 0` false?* Relational operators convert `null` to `0`; `==` has the special null/undefined rule.

TL;DR

Use `===`/`!==`. String-vs-string compares by character code, everything else numerically; `NaN` compares false with everything.

2.7 Logical operators, `??` and `?.`

What it is

Operator	Name	Returns
`a && b`	AND	`a` if `a` is falsy, otherwise `b`
`a || b`	OR	`a` if `a` is truthy, otherwise `b`
`!a`	NOT	`true` if `a` is falsy, else `false` (always a boolean)
`a ?? b`	nullish coalescing (ES2020)	`a` unless it is `null`/`undefined`, then `b`
`a?.b`, `a?.[k]`, `a?.()`	optional chaining (ES2020)	`undefined` if `a` is `null`/`undefined`, otherwise the normal access/call

`&&` and `||` return one of their operands, not necessarily a boolean (*truthy/falsy*: §2.11).

How it works — short-circuit evaluation

- `&&` returns the **first falsy** operand (or the last), `||` the **first truthy** (or the last) — and stop evaluating there.
- So the right side may never run: `user && user.logout()`, `isDebug && console.log(...)`.

Example

```
// code/07_logical_nullish_optional.js (abridged — each output line labels its expression)
const settings = { volume: 0, title: "", theme: null };
console.log("volume || 50:", settings.volume || 50, "| volume ?? 50:", settings.volume ?? 50);

const cfg = { retries: 0, name: "", port: undefined };
cfg.retries ||= 3;    // 0 is falsy -> replaced
cfg.name ??= "app";   // "" is not nullish -> kept
cfg.port ??= 8080;    // undefined -> replaced
let token = "abc"; token &&= token.toUpperCase();
console.log(cfg, token);
```

Output (full run):

```
t && f: false | t || f: true | !f: true
"cat" && "dog": dog
"" && "dog": ""
0 || "default": default
"" || null || 0: 0
!!'text': true | !!0: false
right side evaluated 0 times
volume || 50: 50 | volume ?? 50: 0
title || "Untitled": Untitled | title ?? "Untitled": ""
theme ?? "light": light
user.address?.city: undefined
user.profile?.photo?.url: undefined
user.greet?.(): hi | user.wave?.(): undefined
user.tags?.[0]: undefined
without ?. -> TypeError
{ retries: 3, name: '', port: 8080 } ABC
true || false && false: true
mixing ?? and || without () -> SyntaxError
(null || undefined) ?? 'x': x
```

`||` vs `??` for defaults

`||` falls back on **any falsy** value, discarding legitimate `0`, `""`, `false` (`volume || 50` loses a real volume of 0). `??` falls back **only on** `null`/`undefined`. Use `??` for defaults unless you really mean "any falsy".

Optional chaining `?.`

Reading a property of `null`/`undefined` throws `TypeError`. With `?.`, if the left side is `null`/`undefined` the **whole chain** short-circuits to `undefined`. Forms: `obj?.prop`, `obj?.[expr]`, `fn?.(args)`. You can't assign through it (`obj?.x = 1` is a `SyntaxError`).

Logical assignment (ES2021)

Operator	Equivalent	Assigns when `a` is…
`a ||= b`	`a || (a = b)`	falsy
`a &&= b`	`a && (a = b)`	truthy
`a ??= b`	`a ?? (a = b)`	`null` / `undefined`

They short-circuit: if no assignment is needed, `b` isn't evaluated.

Common mistakes

- `if (x = 5)` assigns (and `5` is truthy); you meant `===`.
- **Overusing** `?.` on values that *must* exist hides bugs behind `undefined`.
- **Mixing** `??` **with** `||`/`&&` **without parentheses** → `SyntaxError`.
- `&&` binds tighter than `||`: `a || b && c` is `a || (b && c)`.

TL;DR

`&&` / `||` short-circuit and return an operand. `??` for defaults (keeps `0`, `""`), `?.` for possibly-missing data, `||=` / `&&=` / `??=` to assign conditionally.

2.8 Bitwise and assignment operators

Bitwise operators

Bitwise operators treat numbers as 32-bit signed integers and work per binary digit (**bit**). Used in low-level code (graphics, compression, hashing, permission flags) and interview puzzles.

Operator	Name	Rule per bit
`&`	AND	1 if both bits are 1
`|`	OR	1 when either bit (or both) is 1
`^`	XOR	1 if the bits differ
`~`	NOT	flips every bit (`~x` equals `-x - 1`)
`<<`	left shift	moves bits left, filling with 0 (×2 per step)
`>>`	sign-propagating right shift	moves bits right, copying the sign bit (÷2, rounding down)
`>>>`	zero-fill right shift	moves bits right, filling with 0 (result is unsigned)

```js
// code/08_bitwise_assignment.js (first half)
let c = 5, d = 2;               // 5 = 101, 2 = 010
console.log("c & d  =", c & d, bin(c & d));
console.log("c | d  =", c | d, bin(c | d));
console.log("c ^ d  =", c ^ d, bin(c ^ d));
console.log("~d     =", ~d);
console.log("c << 1 =", c << 1);
console.log("c >> 1 =", c >> 1, "| -5 >> 1 =", -5 >> 1, "(keeps sign)");
console.log("c >>> 1 =", c >>> 1, "| -5 >>> 1 =", -5 >>> 1, "(fills with 0)");
const READ = 1, WRITE = 2, EXEC = 4;
let perms = READ | WRITE;
console.log("can write?", (perms & WRITE) !== 0, "| can exec?", (perms & EXEC) !== 0);
```

Output:

```
c & d  = 0 000
c | d  = 7 111
c ^ d  = 7 111
~d     = -3
c << 1 = 10
c >> 1 = 2 | -5 >> 1 = -3 (keeps sign)
c >>> 1 = 2 | -5 >>> 1 = 2147483645 (fills with 0)
can write? true | can exec? false
```

`101 & 010` shares no 1-bit → 0; OR and XOR both give `111` = 7. `~2` = `-3` (**two's complement**: negatives stored as "flip all bits, add 1"). The **flags** idiom packs yes/no settings into one number: each flag a power of two, `|` combines, `&` tests (as in Unix `chmod 755`).

Caveat: operands are truncated to 32-bit integers, so bitwise ops break beyond $\pm 2^{31}$ (~2.1 billion). Prefer `Math.trunc` over the old `x | 0` trick.

Assignment operators

Compound assignment combines an operation with `=`: `x += y` means `x = x + y`.

Operator	Same as		Operator	Same as
`x = y`	assign		`x <<= y`	`x = x << y`
`x += y`	`x = x + y`		`x >>= y`	`x = x >> y`
`x -= y`	`x = x - y`		`x >>>= y`	`x = x >>> y`
`x *= y`	`x = x * y`		`x &= y`	`x = x & y`
`x /= y`	`x = x / y`		`x |= y`	`x = x | y`
`x %= y`	`x = x % y`		`x ^= y`	`x = x ^ y`
`x **= y`	`x = x ** y`		`||=` `&&=` `??=`	§2.7

```
// code/08_bitwise_assignment.js (second half, abridged — the file applies every operator in turn)
let p, q;
p = q = 7;                      // right to left: q = 7, then p = (q = 7) which is 7
console.log("p, q:", p, q);     // p, q: 7 7
```

An assignment is an expression whose value is the assigned value — which makes `p = q = 7` work and `if (x = 5)` a silent bug.

2.9 Operator precedence

Precedence decides which operator applies first (`2 + 3 * 4` is `14`); **associativity** orders equal-precedence operators (mostly left-to-right; `**`, assignment and `?:` right-to-left). Highest to lowest (simplified):

Level	Operators
1	`( … )` grouping
2	`.` `[]` `?.` function call `()` `new` with arguments
3	postfix `x++` `x--`
4	`!` `~` unary `+` `-` prefix `++x` `--x` `typeof` `void` `delete` `await`
5	`**` (right-to-left)
6	`*` `/` `%`
7	`+` `-`
8	`<<` `>>` `>>>`
9	`<` `<=` `>` `>=` `in` `instanceof`
10	`==` `!=` `===` `!==`
11–13	`&`, then `^`, then `|`
14	`&&`
15	`||` and `??` (cannot be mixed without parentheses)
16	`? :` (right-to-left)
17	`=` and all compound assignments, `=>` (right-to-left)
18	`,` comma

Don't memorise it — **when in doubt, add parentheses**.

2.10 Numbers in depth

What it is

All ordinary numbers are **64-bit IEEE-754 double-precision floating-point** values ("doubles", like `double` in Java/C). **Floating point** stores significant digits plus an exponent (like 1.23×10^5), giving huge range but approximating most decimal fractions. Consequences:

- about 15–17 significant decimal digits of precision;
- integers exact only up to $2^{53} - 1$ (9,007,199,254,740,991);
- `0.1` can't be stored exactly in binary (like 1/3 in decimal);
- special values `Infinity`, `-Infinity`, `NaN` and a separate `-0`.

The global `Number` holds constants and helpers; `Number(value)` converts (never use `new Number`, §2.4).

Example

`code/09_numbers.js` prints (each line labels the expression it evaluates):

```
0.1 + 0.2 = 0.30000000000000004
0.1 + 0.2 === 0.3 ? false
close enough? true
money in cents: 10 + 20 = 0.3
MAX_VALUE         1.7976931348623157e+308
MIN_VALUE         5e-324
MAX_SAFE_INTEGER  9007199254740991
MIN_SAFE_INTEGER  -9007199254740991
EPSILON           2.220446049250313e-16
POSITIVE_INFINITY Infinity | NEGATIVE_INFINITY -Infinity
MAX_VALUE * 2     Infinity
2**53 + 1         9007199254740992 (not representable!)
isSafeInteger(2**53): false
literals: 1000000 255 15 10 1500 0.002 0.5
toFixed(2): 123.46 | toFixed(0): 123
toPrecision(4): 123.5 | toPrecision(2): 1.2e+2
(255).toString(2): 11111111 | (255).toString(16): ff
(1.005).toFixed(2): 1.00 (binary rounding!)
toLocaleString: 1,234,567.891 | 1.234.567,891
valueOf of wrapper: 7
parseInt('123px'): 123 | parseInt('px123'): NaN
parseInt('ff', 16): 255 | parseInt('101', 2): 5
parseFloat('3.14abc'): 3.14
Number('123px'): NaN | Number(''): 0 | Number(' 42 '): 42
Number(null): 0 | Number(undefined): NaN | Number(true): 1
isInteger(5.0): true | isInteger(5.5): false
isFinite('5'): true | Number.isFinite('5'): false
isNaN('abc'): true | Number.isNaN('abc'): false | Number.isNaN(NaN): true
2n ** 64n = 18446744073709551616n | typeof: bigint
9007199254740993n + 1n = 9007199254740994n
7n / 2n = 3n (truncates)
1n + 1 -> TypeError: Cannot mix BigInt and other types, use explicit conversions
BigInt(10) + 1n = 11n | Number(5n) + 1 = 6
```

Number literals

Form	Example	Value
decimal	`42`, `3.14`, `.5`	
exponent	`1.5e3`, `2e-3`	1500, 0.002
hex / octal / binary	`0xff`, `0o17`, `0b1010`	255, 15, 10
numeric separators (ES2021)	`1_000_000`	1000000 (underscores are ignored, just for readability)
BigInt	`123n`	a `bigint`

`Number` properties

Property	Value	Meaning
`Number.MAX_VALUE`	≈ 1.797e308	largest finite number; beyond it you get `Infinity`
`Number.MIN_VALUE`	5e-324	smallest **positive** number (closest to zero) — *not* the most negative number
`Number.MAX_SAFE_INTEGER`	$2^{53} - 1$	largest integer `n` such that `n` and `n + 1` are both exactly representable
`Number.MIN_SAFE_INTEGER`	$-(2^{53} - 1)$	the negative counterpart
`Number.EPSILON`	≈ 2.22e-16	gap between 1 and the next representable double; a tolerance for comparing decimals near 1
`Number.POSITIVE_INFINITY` / `NEGATIVE_INFINITY`	`Infinity` / `-Infinity`	results of overflow or division by zero
`Number.NaN`	`NaN`	same as the global `NaN`

`Number` methods

Instance methods (on a number value; return strings except `valueOf`):

Method	What it does	Example → result
`toFixed(d)`	round to `d` digits after the decimal point	`(123.456).toFixed(2)` → `"123.46"`
`toPrecision(p)`	round to `p` significant digits (may switch to exponent form)	`(123.456).toPrecision(4)` → `"123.5"`
`toString(radix)`	write in base 2–36	`(255).toString(16)` → `"ff"`
`toLocaleString(locale)`	format for humans (separators, currency) using a locale (e.g. `"en-US"`)	`(1234.5).toLocaleString("en-US")` → `"1,234.5"`
`valueOf()`	the primitive inside a `Number` wrapper	`new Number(7).valueOf()` → `7`

On an integer literal, wrap it: `(255).toString(16)` — `255.toString()` is a syntax error (the dot reads as a decimal point).

Static methods (on `Number`):

Method	What it does	Watch out
`Number.isFinite(v)`	`true` only for finite *numbers*	the global `isFinite("5")` converts first and says `true`
`Number.isInteger(v)`	integer-valued number?	`5.0` is an integer (it's the same value as `5`)
`Number.isSafeInteger(v)`	integer within $\pm(2^{53} - 1)$?	use before trusting large IDs
`Number.isNaN(v)`	is it exactly `NaN`?	the global `isNaN("abc")` converts first and says `true`
`Number.parseFloat(s)`	read a decimal from the *start* of a string	stops at the first invalid character
`Number.parseInt(s, radix)`	read an integer from the *start* of a string in base `radix`	**always pass the radix** (usually 10)

`parseInt` / `parseFloat` read a leading number and ignore the rest (`"123px"` → `123`, good for CSS values); `Number()` needs the *whole* trimmed string numeric, else `NaN`, and treats `""` as `0`.

`NaN` and `Infinity`

- `NaN` results from failed numeric operations (`Number("abc")`, `0 / 0`, `Math.sqrt(-1)`), spreads through arithmetic, and is the only value not equal to itself. Use `Number.isNaN`.
- `Infinity` results from overflow or division by zero. `Number.isFinite(x)` rejects `±Infinity` and `NaN` at once.

Floating-point precision — what to do about it

1. **Never compare decimals with `===`**: `Math.abs(a - b) < 1e-9` (or `Number.EPSILON` near 1).
2. **Money in integer cents** (store 1999, show 19.99) or a decimal library — payment APIs like Stripe take the smallest currency unit for this reason.
3. **Round only for display** (`toFixed`, `Intl.NumberFormat`/`toLocaleString`). `toFixed` rounds the *binary* value: `(1.005).toFixed(2)` is `"1.00"` (stored as 1.00499999…).
4. **Integers beyond 2^{53}** (DB IDs, X post IDs, nanosecond timestamps) lose precision silently — keep them as **strings** or `BigInt`.

`BigInt`

`bigint` (ES2020): whole numbers of any size, written `10n` or `BigInt(10)`. **No mixing** with `number` in arithmetic (`TypeError`) — convert explicitly; division truncates (`7n / 2n` → `3n`); `Math` rejects BigInts; `JSON.stringify` can't serialise them without help. Use for big IDs, cryptography, exact large-integer maths; `number` otherwise (faster).

In the interview

"Why is `0.1 + 0.2 !== 0.3`?" Numbers are IEEE-754 binary doubles; 0.1 and 0.2 have no exact binary form, so each is the nearest double and their sum is the double just above 0.3 (0.30000000000000004). Compare with a tolerance or use integer units like cents.

- *`Number.MAX_SAFE_INTEGER`?* $2^{53} - 1$; above it not every integer is representable (`2**53 + 1 === 2**53`). Big IDs → strings or BigInt.
- *`isNaN` vs `Number.isNaN`?* The global converts first (`isNaN("abc")` true); `Number.isNaN` is true only for `NaN`.
- *`parseInt("08")`?* `8` in modern engines — still pass the radix. `parseInt(0.0000005)` is `5` (the number becomes the string `"5e-7"` first).

Self-check

1. `Number("")` vs `parseInt("")`? → `0` vs `NaN`.
2. `Number.MIN_VALUE`? → Smallest *positive* double, 5e-324.
3. `typeof NaN`? → `"number"`.
4. `10n + 5`? → `TypeError`.

TL;DR

One `number` type (64-bit float): exact integers to $2^{53} - 1$, inexact decimals, `NaN`, `±Infinity`. `Number.isNaN/isFinite/isInteger`; `toFixed` for display; cents for money; `BigInt` for huge integers.

2.11 Booleans, truthy and falsy

What it is

`boolean` has two values, `true` and `false`; conditions (`if`, `while`, `? :`, Chapter 3) consume them. Any value can be a condition: **falsy** values become `false`, **everything else is truthy**.

Falsy values (the complete list)
`false`
`0`, `-0` and `0n` (BigInt zero)
`""` — the empty string, whichever quotes are used
`null`
`undefined`
`NaN`

(Plus the legacy browser oddity `document.all`.) Surprising **truthy** values: `"0"`, `"false"`, `" "`, `[]`, `{}`, `-1`, `Infinity`, every function, every wrapper object — including `new Boolean(false)`.

Example

```
// code/10_boolean_truthiness.js (abridged — each output line labels its expression)
if (false || 0 || "" || null || undefined || NaN) {
  console.log("never printed");
} else {
  console.log("All values are falsy.");
}
const wrapped = new Boolean(false);
console.log("if (new Boolean(false)) runs?", wrapped ? "yes (object is truthy)" : "no");
```

Output:

```
falsy: false->false  0->false  -0->false  0n->false  ""->false  null->false  undefined->false  NaN->false
truthy: true
All values are falsy.
Boolean("hi"): true | !!"": false | !!42: true
typeof wrapped: object | valueOf(): false
if (new Boolean(false)) runs? yes (object is truthy)
new Boolean(true).valueOf(): true | toString(): true
isAdult: true boolean
true + true = 2 | [true, false, true].filter(Boolean).length = 2
```

(All six `||` operands are falsy, so it returns the last, `NaN`, and the `else` runs.)

Converting to boolean

- **`Boolean(value)`** — explicit; **`!!value`** — double NOT shorthand; **`array.filter(Boolean)`** — drops falsy entries.

The `Boolean` wrapper object

`new Boolean(x)` is an object (methods: `valueOf()`, `toString()`) and therefore *always truthy*, even `new Boolean(false)`. Use `Boolean(x)` without `new`.

Common mistakes

- **`if (arr)` for "non-empty"** — `[]` is truthy; use `arr.length > 0`.
- **`if (count)` when `0` is valid** — use `count !== undefined` or `??`.
- **`if (x == true)`** — `true` becomes `1`, so `"yes" == true` is false. Use `if (x)` or `x === true`.
- **`"false"` from a form/URL** is truthy — compare `value === "true"`.

In the interview

"List the falsy values." `false`, `0`, `-0`, `0n`, `""`, `null`, `undefined`, `NaN` (plus legacy `document.all`). Everything else — including `"0"`, `[]`, `{}`, `new Boolean(false)` — is truthy.

- *`!!"false"`?* `true` — non-empty string.
- *Why is `new Boolean(false)` truthy?* All objects are truthy.

TL;DR

Eight falsy values; everything else truthy. Convert with `Boolean(x)` or `!!x`; never `new Boolean`.

2.12 Type conversion summary

Type conversion (**coercion**) turns a value of one type into another — **explicit** when you ask (`Number(x)`, `String(x)`, `Boolean(x)`), **implicit** when an operator does it (`"3" * 2`, `if (x)`, `x == y`).

```
// code/11_conversions.js (the table loop)
const values = ["42", "", " ", "4px", "0x1f", null, undefined, true, false, [], [7], [1, 2], {}];
for (const v of values) {
  console.log(show(v).padEnd(10), String(Number(v)).padEnd(10), JSON.stringify(String(v)).padEnd(18), Boolean(v));
}
```

Output:

```
value      Number()   String()           Boolean()
"42"       42         "42"               true
""         0          ""                 false
" "        0          " "                true
"4px"      NaN        "4px"              true
"0x1f"     31         "0x1f"             true
null       0          "null"             false
undefined  NaN        "undefined"        false
true       1          "true"             true
false      0          "false"            false
[]         0          ""                 true
[7]        7          "7"                true
[1,2]      NaN        "1,2"              true
{}         NaN        "[object Object]"  true
parseInt vs Number on '08.5kg': 8 NaN
String(123) + 1 = 1231 | `${123}`: 123
```

1. **To number:** whitespace-only → `0`; numeric strings (incl. `0x…`, `0b…`, `0o…`) → the number; other text → `NaN`; `null` → `0`, `undefined` → `NaN`, booleans → `1`/`0`; objects → primitive first (arrays → joined string).
2. **To string:** arrays join with commas (`[]` → `""`); plain objects → `"[object Object]"` (use `JSON.stringify` to see contents).
3. **To boolean:** the §2.11 falsy list → `false`; else `true`.

Context	Converts to
`if`, `while`, `? :`, `!`, `&&`, `||`	boolean
`-` `*` `/` `%` `**`, unary `+`/`-`, `<` `>` (unless both strings)	number
`+` when either side is a string; template literals `` `${x}` ``	string
`==`	follows the §2.6 algorithm

Prefer **explicit** conversions (`Number(input.value)`) and `===`.

Chapter 2 TL;DR

- **Declare** with `const` (default) or `let`; avoid `var` (function-scoped, redeclarable, hoisted as `undefined`). `let`/`const` are block-scoped and throw in the **TDZ**.
- **Types:** seven primitives (`number`, `bigint`, `string`, `boolean`, `undefined`, `null`, `symbol`) + objects. `typeof null` is `"object"`; use `Array.isArray`.
- **Primitives** are immutable, copied by value; **objects** copied and compared by reference.

- **Arithmetic:** `/` is floating point; `%` keeps the left sign; `+` concatenates if either side is a string; postfix `++` returns the old value.
- **Equality:** `===`; `==` coerces non-transitively (`x == null` is the exception). `Object.is` fixes `NaN` and `-0`.
- **Logic:** `&&`/`||` short-circuit and return operands; `??` defaults only on `null`/`undefined`; `?.` reads possibly-missing data; `||=`, `&&=`, `??=` assign conditionally.
- **Numbers:** IEEE-754 doubles — `0.1 + 0.2 !== 0.3`, safe integers to $2^{53} - 1$, `NaN`/`Infinity`; `Number.isNaN`, `toFixed` for display, cents for money, `BigInt` for huge integers; `parseInt(s, 10)` vs `Number(s)` differ on trailing text.
- **Truthiness:** falsy = `false, 0, -0, 0n, "", null, undefined, NaN`; everything else (incl. `[]`, `"0"`, `new Boolean(false)`) truthy.

Next: Chapter 3 — Control Flow and Loops.